

# Assignment 1: MVC Design Pattern and Its Variants

## Overview

The Model-View-Controller (MVC) design pattern divides an application into three interconnected components:

- **Model:** Manages the data and business logic.
- **View:** Handles the presentation layer and user interface.
- **Controller:** Processes user input and interacts with the Model and View.

This separation facilitates modular development, easier testing, and maintenance.

## Variants of MVC

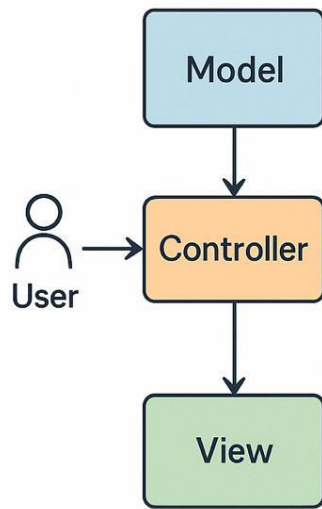
**Model-View-Presenter (MVP):**

- **Presenter:** Acts as an intermediary between the View and Model, handling all presentation logic.
- **Use Case:** Suitable for applications requiring clear separation between UI and business logic, such as Windows Forms applications.

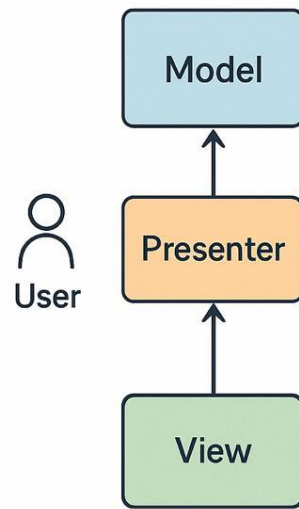
**Model-View-ViewModel (MVVM):**

- **ViewModel:** Exposes data and commands from the Model to the View, facilitating data binding.
- **Use Case:** Ideal for applications using data-binding capabilities, like WPF or Angular applications.

## MVC



## MVP



## MVVM

